DO NOW — RECALL

# Linked lists using objects

Name: _____ Class: _____ Date: _____

**1. What makes a heuristic admissible?**

**2. What does a partition do to the pivot?**

**3. What does the head reference identify?**

**4. Why must a new node's Next be set before replacing an existing Next link?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

It never overestimates the true cheapest remaining cost.

### 2. Model answer

Places it at its final index with smaller/equal values to its left.

### 3. Model answer

The first node in a linked list.

### 4. Model answer

Otherwise the original successor can become unreachable and part of the list may be lost.

### Vocabulary — know these terms

node — An object containing data and a reference to another node.

head — Reference to the first node.

Next — Link to the successor node.

null — No object is referenced.

traversal — Visit nodes successively by following Next references.

NEW LEARNING

# Read, annotate and explain

### Build a chain

Each Node stores a Value and a Next reference; the final Next is null.

### Preserve the successor

Construct 15 with Next set to the old 20 node *before* changing 10.Next.

### Finish the insertion

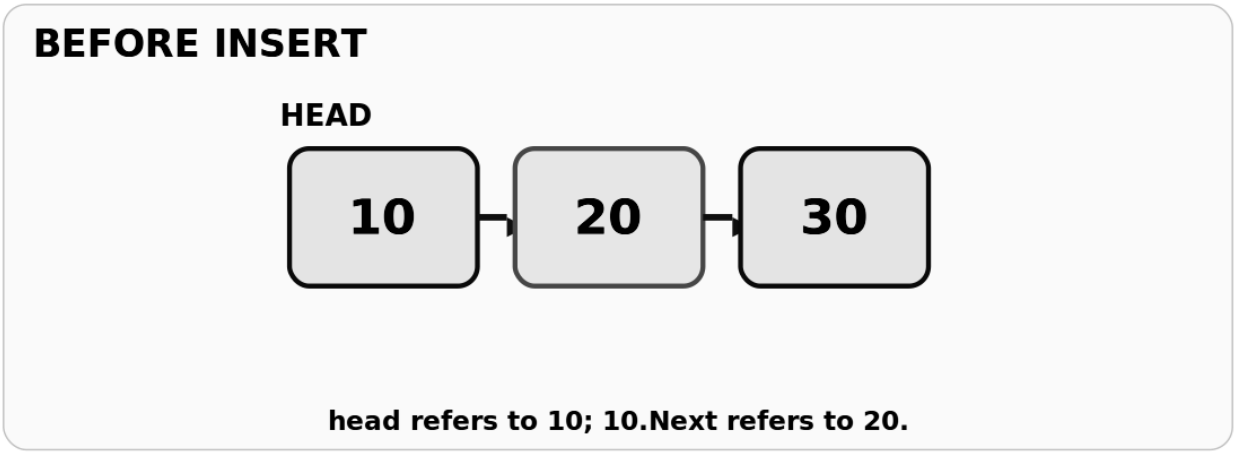
Set 10.Next to 15. Now traversal yields 10→15→20→30.

### Explain one step in your own words, using the correct technical terms.

# New Learning · original lesson diagrams

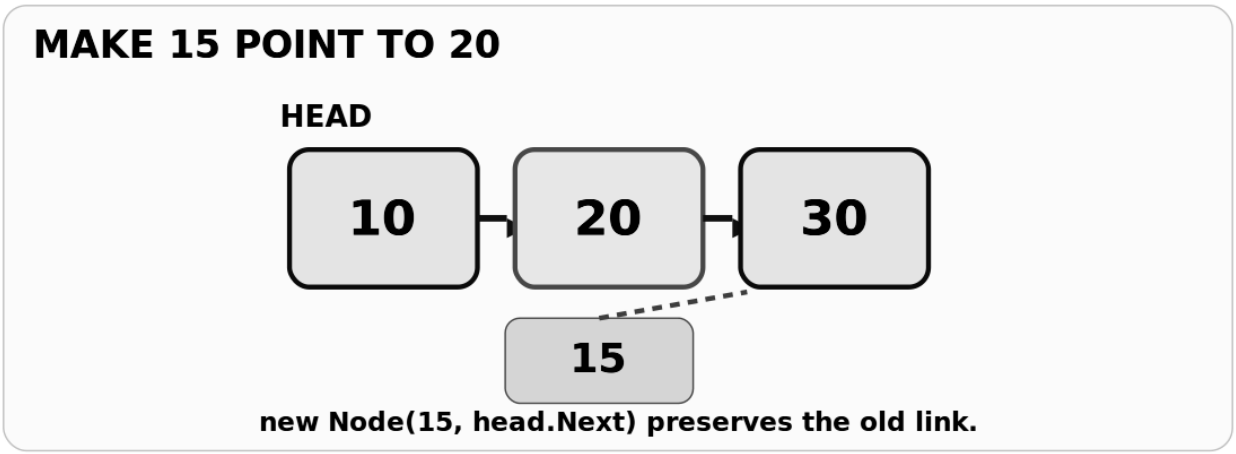
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




BEFORE INSERT: 10 to 20 to 30. head refers to 10; 10.Next refers to 20.

### New Learning

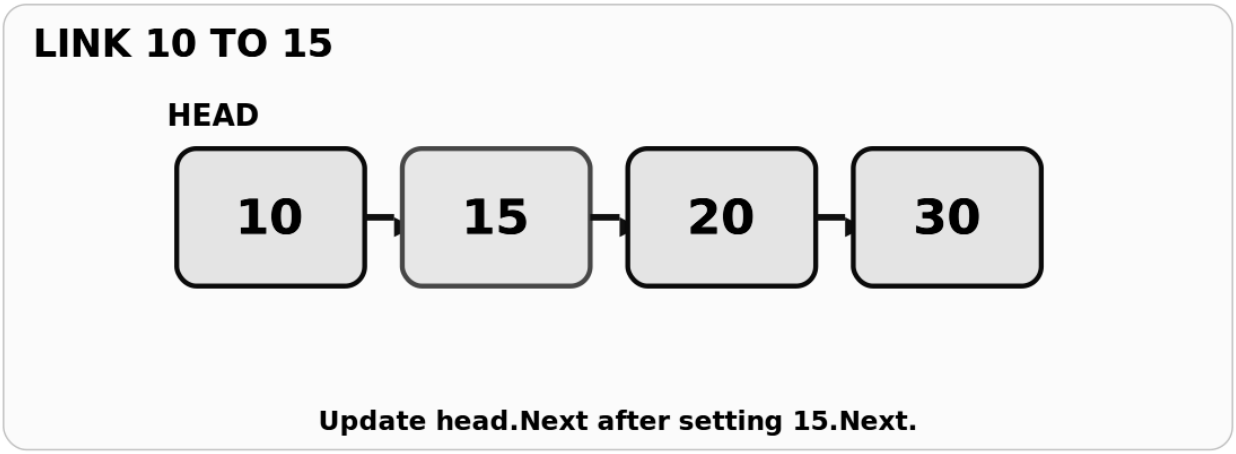



MAKE 15 POINT TO 20: 10 to 20 to 30. new Node(15, head.Next) preserves the old link.

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




LINK 10 TO 15: 10 to 15 to 20 to 30. Update head.Next after setting 15.Next.

I DO — WORKED EXAMPLE

# Follow the worked method

### Insert node 7 between 4 and 9 in 4 → 9 → 12.

### Predict first

Set newNode.Next to the existing successor 9.

### Model the reasoning

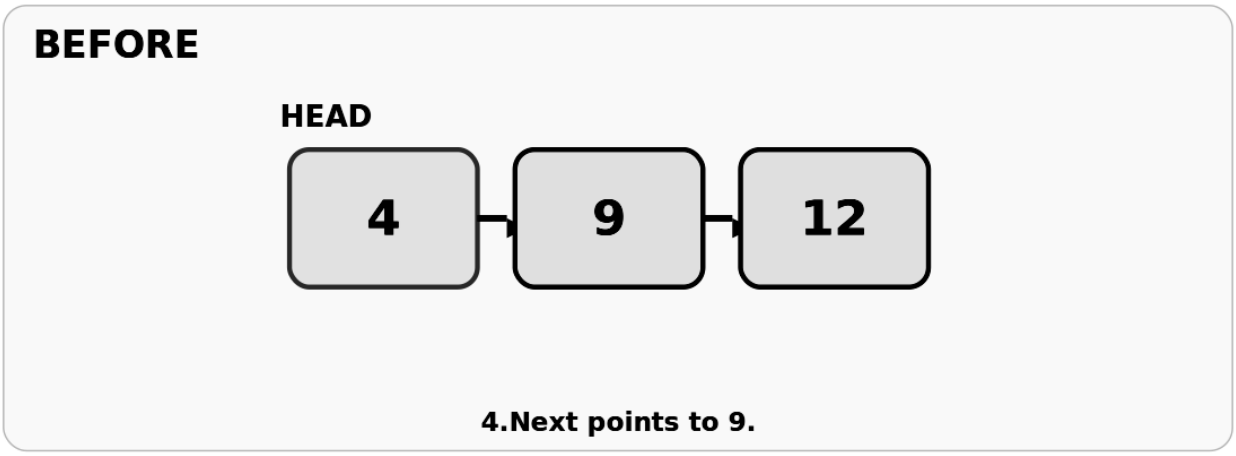
Update node4.Next to reference the new 7 node. The 9→12 remainder is preserved.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Insert node 7 between 4 and 9 in 4 → 9 → 12.

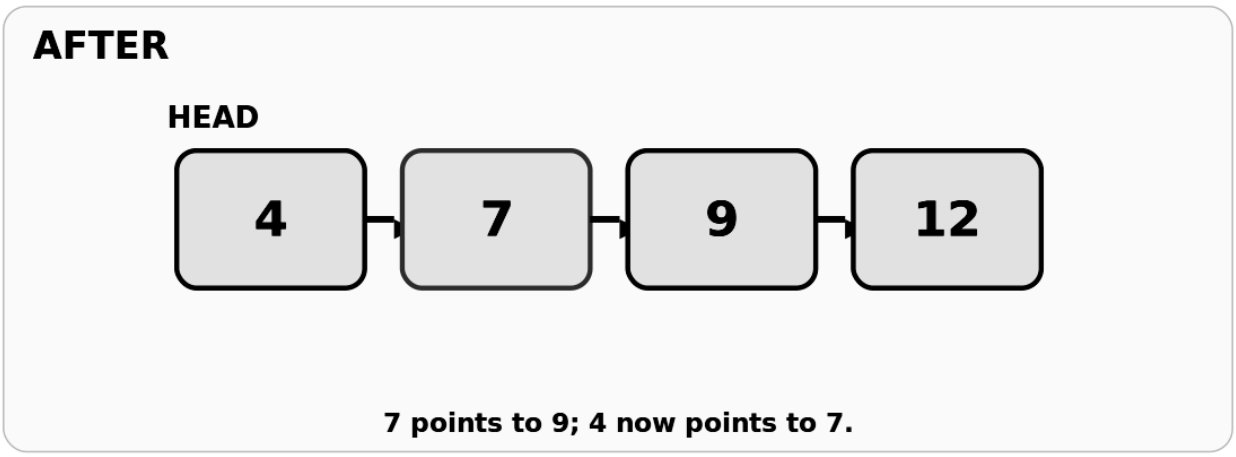
Set newNode.Next to the existing successor 9.




BEFORE: 4 to 9 to 12. 4.Next points to 9.

### Insert node 7 between 4 and 9 in 4 → 9 → 12.

Set newNode.Next to the existing successor 9.




AFTER: 4 to 7 to 9 to 12. 7 points to 9; 4 now points to 7.

WE DO — GUIDED PRACTICE

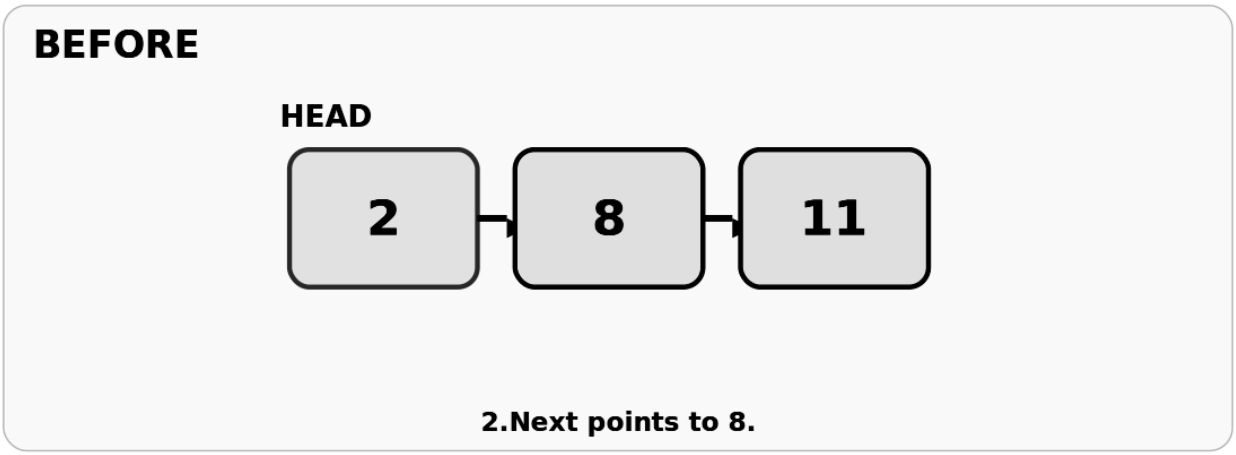
# Try → Discuss → Improve

# We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Insert 6 after node 2 in 2 → 8 → 11.

First give the new node a reference to 8, the former successor of 2.




BEFORE: 2 to 8 to 11. 2.Next points to 8.

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Predict first First give the new node a reference to 8, the former successor of 2.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

For 3→5→8, identify head, 5.Next.Value and 8.Next.

### Question 2

Write two C# statements to insert a node with value 12 after node8 in 8→20→30. Give the new traversal.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Describe how to delete node 20 from 10→20→30 if you hold a reference to node10. Explain the complexity of reaching the deletion point and changing the link.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Inserting a linked-list node always requires shifting the stored values of later nodes.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

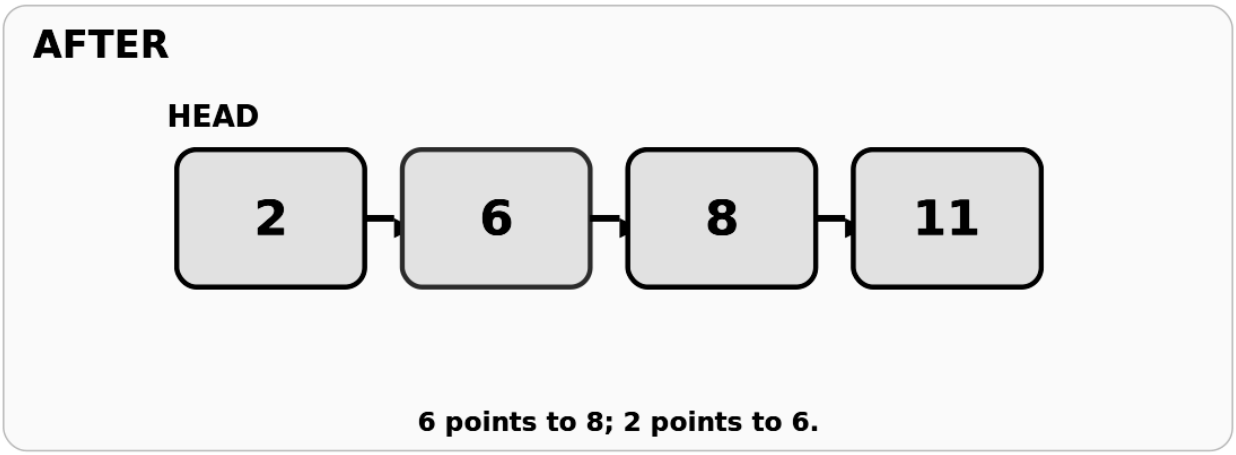
Next step: revisit the model or a relevant revision-guide section if needed.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Insert 6 after node 2 in 2 → 8 → 11.

First give the new node a reference to 8, the former successor of 2.




AFTER: 2 to 6 to 8 to 11. 6 points to 8; 2 points to 6.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Model the reasoning

Then change node2.Next to reference 6; the final chain is 2→6→8→11.

IMPROVE: Explain one correction to your own reasoning.